Matrix factorization: ALS and beyond

A compact companion to Seminar 04. All symbols and objectives match the notebook.

1. Observed ratings and dimensions

Let R have m users and n movies. Only pairs in Omega are observed. U has m rows and d latent dimensions; V has n rows and the same d dimensions. An unobserved R[u,i] is unknown, not zero.

```
R: m x n    U: m x d    V: n x d
predicted rating: r_hat[u,i] = dot(U[u], V[i])
```

2. Explicit ALS objective

```
L(U,V) = sum_(u,i in Omega) (r[u,i] - dot(U[u],V[i]))^2
         + lambda * sum_u ||U[u]||^2 + lambda * sum_i ||V[i]||^2
```

The objective is not jointly convex in U and V. Holding either matrix fixed produces independent ridge regressions. Alternating these two exact block updates gives ALS; it need not find the global minimum.

3. One user update

Let I(u) be movies rated by user u. Stack their factor rows into V_u and their observed ratings into r_u. Differentiating the user-specific ridge objective gives the normal equation:

```
(V_u^T V_u + lambda I_d) U[u] = V_u^T r_u
```

Solve the d-by-d linear system. Do not form the inverse explicitly. For one movie i, swap users and movies:

```
(U_i^T U_i + lambda I_d) V[i] = U_i^T r_i
```

An empty history has no rating evidence. A zero factor (or an explicit fallback) is more meaningful than leaving a random initialized factor.

Related factorization models

4. Similar notation, different fitting procedures

Method	Target and training	What to remember
Explicit ALS	Observed rating squared error; alternating ridge solves.	Basic notebook version has no bias terms.
SVD-style MF	Observed rating error; SGD on user/item factors and biases.	Its name does not mean a literal matrix SVD.
SVD++	Observed rating error; SGD plus factors for items in the user history.	History gives implicit evidence even when predicting explicit stars.
Truncated matrix SVD	Low-rank approximation of a complete numerical matrix.	Zero-filling missing ratings changes the task.

5. SVD and SVD++ prediction

```
SVD:   r_hat[u,i] = mu + b_u + b_i + dot(p_u, q_i)
SVD++: r_hat[u,i] = mu + b_u + b_i
                    + dot(q_i, p_u + sum_(j in N(u)) y_j / sqrt(|N(u)|))
```

Here mu is the global mean; b_u and b_i are user and item biases. N(u) is the set of movies in the training history, and y_j is an additional history factor. The SVD++ history term can improve sparse-user predictions, but costs more to train.

6. Why zero-filled SVD is not a fair ALS baseline

Classical SVD treats every matrix entry as supplied. A sparse rating matrix materialized as zeros tells it that every missing user-movie pair has a true rating of zero. Observed-only ALS and SVD-style factorization fit only known ratings. Singular values of the zero-filled matrix are useful as a structural picture, but the objectives differ.

The notebook therefore compares ALS, SVD-style factorization and SVD++ on the same observed train interactions, then evaluates predictions for the same warm test movies. All three implementations run without extra Colab packages.

Implicit feedback and evaluation

7. Implicit ALS

Clicks, views and plays are interaction counts, not explicit 1-5 star judgments. Define preference p[u,i] as 1 for an observed positive interaction and 0 otherwise. Define confidence c[u,i] = 1 + alpha * r[u,i] for observed counts and 1 for unobserved pairs.

```
L_implicit(U,V) = sum_(all u,i) c[u,i] * (p[u,i] - dot(U[u],V[i]))^2
                  + lambda * (sum_u ||U[u]||^2 + sum_i ||V[i]||^2)
```

The sum includes unobserved pairs with low confidence; they are not known dislikes. The fast implicit ALS update exploits the shared item-factor Gram matrix rather than materializing the full dense matrix. Its scores rank likely preferences and are not predicted star ratings.

8. A trustworthy offline comparison

Stage	Rule
Split	For each user: earlier ratings to train, penultimate to validation, latest to test. Break timestamp ties deterministically.
Tune	Choose ALS settings from validation. Refit the chosen model on train plus validation. Read test once.
Ranking labels	A held-out rating of at least 4 stars is relevant; 1-3 stars are not hits. Report the number of eligible users.
Candidates	Use the same warm-item universe for every model, exclude items already rated in training, and report coverage.
Metrics	RMSE/MAE for rating prediction; HitRate@K and NDCG@K for positive-item ranking. Include a simple baseline.

Interpret results within this protocol. Offline held-out scores do not estimate online impact, and warm-item evaluation says little about new movies or users.

Data: GroupLens MovieLens 1M, grouplens.org/datasets/movielens/1m/